

MTH 446: Lecture # 16

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Lecture Span

1. Complex Logs

Complex logs

Find $2 \log(-1 + i)$. Note that

$$-1 + i = \sqrt{2} \left(-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right) \quad (1)$$

$$= \sqrt{2} (\cos(3\pi/4) + i \sin(3\pi/4)) \quad (2)$$

$$= \sqrt{2} \exp(i3\pi/4) \quad (3)$$

Therefore,

$$\arg(-1 + i) = 3\pi/4 \quad (4)$$

Therefore,

$$\log(-1 + i) = \ln(\sqrt{2}) + i \frac{3\pi}{4} \quad (5)$$

Thus,

$$2 \log(-1 + i) = \ln(2) + i \frac{3\pi}{2} \quad (6)$$

Note, $\log(z^2) = 2 \log(z)$.

Analyticity of the single-valued branch of $\log(z)$

Consider the single-valued branch of $\log(z)$, defined as

$$\log(z) = \ln(r) + i\theta, \quad \alpha < \theta < 2\pi + \alpha \quad (7)$$

Then, the domain of this log is $\mathbb{C}/\text{branch cut}$. Thus, $\log : D_\alpha \rightarrow \mathbb{C}$. Then,

$$\log(z) = u(r, \theta) + iv(r, \theta) \quad (8)$$

Moreover, $u = \ln(r)$, and $v(r, \theta) = \theta$. Thus,

$$u_r = 1/r \quad (9)$$

$$u_\theta = 0 \quad (10)$$

$$v_r = 0 \quad (11)$$

$$v_\theta = 1 \quad (12)$$

The CR equations in Polar are

$$u_r = \frac{1}{r}v_\theta, \quad v_r = -\frac{1}{r}u_\theta \quad (13)$$

The first equation is satisfied. The second equation is also satisfied. Therefore, $\log(z)$ is complex differentiable except for the branch cut.

Derivative of $\log(z)$

Let $\log(z) = \ln r + i\theta$. Then recall that

$$f'(z) = -\frac{i}{z}(u_\theta + iv_\theta) \quad (14)$$

Therefore,

$$f'(z) = -\frac{i}{z}(0 + i) = \frac{1}{z} \quad (15)$$

If $z = 0$ is called the branch point of $\log(z)$ (the point at which all the branch cuts intersect).

Bernoulli's Sophism

Bernoulli showed that $\log(z) = \log(-z)$. But this isn't a proof, so it's a sophism.

Identities

Define,

$$c \log(z) = c [\ln r + i(\theta + 2\pi k)] \quad (16)$$

We can prove the following:

1. $z^n = \exp(n \log z)$
2. $z^{1/n} = \exp(\frac{1}{n} \log z)$

The power function

Note that $z^n = \exp(n \log(z))$. This motivates the following definition:

$$z^c = \exp(c \log z), \quad c \in \mathbb{R} \quad (17)$$

Therefore, z^c is a multiple valued function. The principle value is

$$P.V z^c = \exp(c \text{Log}(z)) \quad (18)$$

Example,

$$i^i = \exp(i \log(i)) \quad (19)$$

Note that $i = \exp(i\pi/2)$. Then,

$$\exp(i(\ln(1) + i(\pi/2 + 2\pi k))) = \exp\left(-\frac{4k+1}{2}\pi\right), k \in \mathbb{Z} \quad (20)$$

Thus,

$$i^i = \exp(i \text{Log}(i)) = \exp(-\pi/2) \quad (21)$$

Example

Let $\alpha = 3\pi/4$. Then, use the single valued log to evaluate z^i . Then,

$$i^i = \exp(i \log(i)) = \exp(i\theta), \quad \theta \in (\alpha, 2\pi + \alpha) \quad (22)$$

Therefore,

$$\theta = 5\pi/2 \quad (23)$$

is the value we want.

Evaluate

$$(1 - i)^{4i} \quad (24)$$

This is the same as:

$$\exp(4i \log(1 - i)) \quad (25)$$